

Foundations of Ensemble learning

[Ensemble learning]

1.0 Content Block

Boosting involves training successive models by emphasizing training data mis-classified by previously learned models. Initially, all data (D_1) has equal weight and is used to learn a base model M_1 . The examples mis-classified by M_1 are assigned a weight greater than correctly classified examples. This boosted data (D_2) is used to train a second base model M_2 , and so on. Inference is done by voting. In some cases, boosting has yielded better accuracy than bagging, but tends to over-fit more. The most common implementation of boosting is Adaboost, but some newer algorithms are reported to achieve better results.

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The Bayes optimal classifier is a classification technique. It is an ensemble of all the hypotheses in the hypothesis space. On average, no other ensemble can outperform it. The Naive Bayes classifier is a version of this that assumes that the data is conditionally independent on the class and makes the computation more feasible. Each hypothesis is given a vote proportional to the likelihood that the training dataset would be sampled from a system if that hypothesis were true. To facilitate training data of finite size, the vote of each hypothesis is also multiplied by the prior probability of that hypothesis. The Bayes optimal classifier can be expressed with the following equation:

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Intrusion detection An intrusion detection system monitors computer network or computer systems to identify intruder codes like an anomaly detection process. Ensemble learning successfully aids such monitoring systems to reduce their total error.

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where e^k is the cost function of the k^{th} classifier, q^k is the probability of the k^{th} classifier, p is the true probability that we need to estimate and λ is a parameter between 0 and 1 that define the diversity that we would like to establish. When $\lambda = 0$ we want each classifier to do its best regardless of the ensemble and when $\lambda = 1$ we would like the classifier to be as diverse as possible.

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Distributed denial of service Distributed denial of service is one of the most threatening cyber-attacks that may happen to an internet service provider. By combining the output of single classifiers, ensemble classifiers reduce the total error of detecting and discriminating such attacks from legitimate flash crowds.

6.0 Content Block

In statistics and machine learning, ensemble methods use multiple learning algorithms to obtain better predictive performance than could be obtained from any of the constituent learning algorithms alone. Unlike a statistical ensemble in statistical mechanics, which is usually infinite, a machine learning ensemble consists of only a concrete finite set of alternative models, but typically allows for much more flexible structure to exist among those alternatives.

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Bayesian model averaging Bayesian model averaging (BMA) makes predictions by averaging the predictions of models weighted by their posterior probabilities given the data. BMA is known to generally give better answers than a single model, obtained, e.g., via stepwise regression, especially where very different models have nearly identical performance in the training set but may otherwise perform quite differently. The question with any use of Bayes' theorem is the prior, i.e., the probability (perhaps subjective) that each model is the best to use for a given purpose. Conceptually, BMA can be used with any prior. R packages ensembleBMA and BMA use the prior implied by the Bayesian information criterion, (BIC), following Raftery (1995). R package BAS supports the use of the priors implied by Akaike information criterion (AIC) and other criteria over the alternative models as well as priors over the coefficients. The difference between BIC and AIC is the strength of preference for parsimony. BIC's penalty for model complexity is $\ln(n)k$, while AIC's is $2k$. Large-sample asymptotic theory establishes that if there is a best model, then with increasing sample sizes, BIC is strongly consistent, i.e., will almost certainly find it, while AIC may not, because AIC may continue to place excessive posterior probability on models that are more complicated than they need to be. On the other hand, AIC and AICc are asymptotically "efficient" (i.e., minimum mean square prediction error), while BIC is not. Haussler et al. (1994) showed that when BMA is used for classification, its expected error is at most twice the expected error of the Bayes optimal classifier. Burnham and Anderson (1998, 2002) contributed greatly to introducing a wider audience to the basic ideas of Bayesian model averaging and popularizing the methodology. The availability of software, including other free open-source packages for R beyond those mentioned above, helped make the methods accessible to a wider audience.

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$$e^k = H(p, q^k) - \frac{\lambda}{K} \sum_{j \neq k} H(q^j, q^k) \quad \{ \text{displaystyle } e^k = H(p, q^k) - \frac{\lambda}{K} \sum_{j \neq k} H(q^j, q^k) \}$$

9.0 Content Block

Bayesian model combination Bayesian model combination (BMC) is an algorithmic correction to Bayesian model averaging (BMA). Instead of sampling each model in the ensemble individually, it samples from the space of possible ensembles (with model weights drawn randomly from a Dirichlet distribution having uniform parameters). This modification overcomes the tendency of BMA to converge toward giving all the weight to a single model. Although BMC is somewhat more computationally expensive than BMA, it tends to yield dramatically better results. BMC has been shown to be better on average (with statistical significance) than BMA and bagging. Use of Bayes' law to compute model weights requires computing the probability of the data given each model. Typically, none of the models in the ensemble are exactly the distribution from which the training data were generated, so all of them correctly receive a value close to zero for this term. This would work well if the ensemble were big enough to sample the entire model-space, but this is rarely possible. Consequently, each pattern in the training data will cause the ensemble weight to shift toward the model in the ensemble that is closest to the distribution of the training data. It essentially reduces to an unnecessarily complex method for doing model selection. The possible weightings for an ensemble can be visualized as lying on a simplex. At each vertex of the simplex, all of the weight is given to a single model in the ensemble. BMA converges toward the vertex that is closest to the distribution of the training data. By contrast, BMC converges toward the point where this distribution projects onto the simplex. In other words, instead of selecting the one model that is closest to the generating distribution, it seeks the combination of models that is closest to the generating distribution. The results from BMA can often be approximated by using cross-validation to select the best model from a bucket of models. Likewise, the results from BMC may be approximated by using cross-validation to select the best ensemble combination from a random sampling of possible weightings.